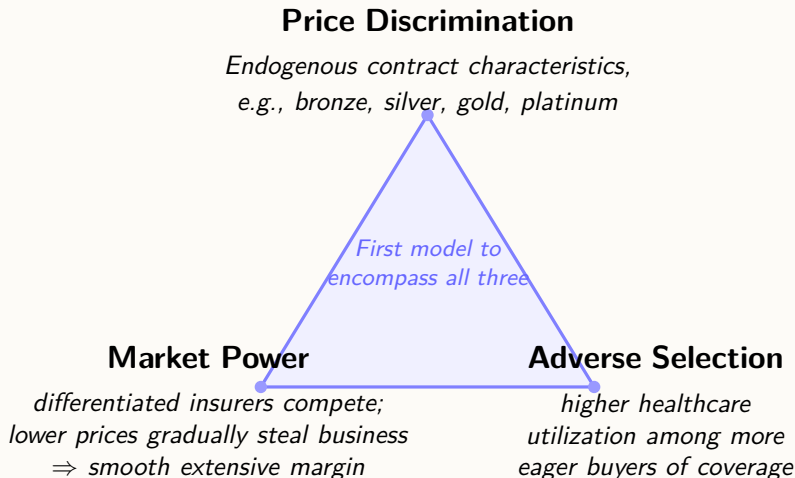


# **Price Discrimination and Market Power in Insurance Markets**

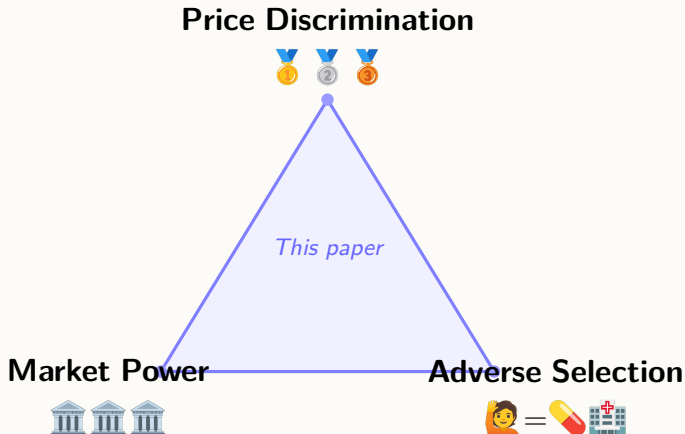
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Oxford April 2026

# Price Discrimination + Market Power in Insurance



# Price Discrimination + Market Power in Insurance



# Preview of Results

## Finding 1: Cross-subsidization

- ▶ Insurers optimally sell comprehensive coverage *at a loss*
- ▶ Financed by profit on intermediate contracts

## Finding 2: Extensive margin exacerbates distortions

- ▶ With private values, the extensive margin *attenuates* monopoly distortions  
(Rochet & Stole 2002, Yang & Ye 2008)
- ▶ With adverse selection, the extensive margin ***exacerbates*** distortions

## Finding 3: Competition exacerbates distortions at the top

- ▶  $\pi(\bar{\theta}) < 0 \Rightarrow$  *lemon-dropping*, not business stealing
- ▶ Insurers raise premia to shed high-risk buyers onto rivals
- ▶ At the bottom, competition *improves* inclusion

**Technical:** Existence and uniqueness of pure-strategy equilibrium under sufficient horizontal differentiation

## Related Literature

### Methodological starting point:

Rochet & Stole (2002), Yang & Ye (2008)

Imperfect competition + nonlinear pricing under *private values*

⇒ We embed adverse selection, reversing several of their key insights

### Market power + adverse selection (single-contract menus):

Mahoney & Weyl (2017), Parlasca (2021), Kong, Layton & Shepard (2025)

### Price discrimination + adverse selection (no market power):

Perfect competition: Rothschild & Stiglitz (1976), Riley (1979),  
Azevedo & Gottlieb (2017), Luz, Gottardi & Moreira (2023)

Pure monopoly: Stiglitz (1977), Chade & Schlee (2012, 2020)

Monopoly vs. competition: Gottlieb & Moreira (WP)

### Imperfect competition + screening (binary types):

Lester, Shourideh, Venkateswaran & Zetlin-Jones (2019)

⇒ mixed strategies; our continuum-type model yields pure-strategy eq.

Set-up

# Private Health Insurance

## Vertically differentiated demand:

Mean health expenditure  $\theta \sim F$  with  $\theta \in [\underline{\theta}, \bar{\theta}]$

High  $\theta$  types:  =  

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Premium  $p \in \mathbb{R}_+$  and coverage  $q \in [0, 1 - \underline{Q}]$

$q$  represents the fraction of expenses reimbursed by the insurer

Total coverage:  $Q \equiv \underline{Q} + q \in [\underline{Q}, 1]$



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Total coverage:  $Q \equiv \underline{Q} + q \in [\underline{Q}, 1]$

**Direct Mechanism:**  $(Q(\theta), p(\theta))$

**Buyers can opt out:** Derive utility from basic coverage  $\underline{Q}$

# Tractable Utility

**Buyers are risk-averse:** *CARA utility + normal health shocks*

Certainty equivalent quadratic in  $Q$  and linear in  $\theta$  and premium  $p$ :

$$U(Q; \theta) - p = \theta Q - \frac{\gamma}{2}(1 - Q)^2 - p$$

where  $\gamma = \text{CARA coefficient} \times \text{variance of health shocks}$

Derivation from CARA utility

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Derivation from CARA utility

**Insurers are risk-neutral:** Profit is premium minus expected cost:

$$p - \theta q$$

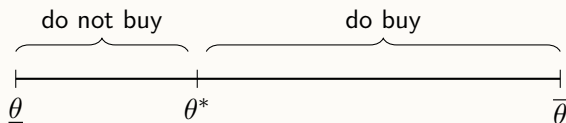
Adverse selection: high- $\theta$  buyers are costlier to insure

## Benchmark: Pure Monopoly

Monopolist maximizes virtual surplus (Myerson 1981):

$$\int_{\theta^*}^{\bar{\theta}} \left\{ \underbrace{\theta \underline{Q} - \frac{\gamma}{2}(1 - Q(\theta))^2 - u(\underline{Q}; \theta)}_{\text{surplus}} - \underbrace{(Q(\theta) - \underline{Q}) \frac{1 - F(\theta)}{f(\theta)}}_{\text{information rent}} \right\} f(\theta) d\theta$$

No extensive margin:

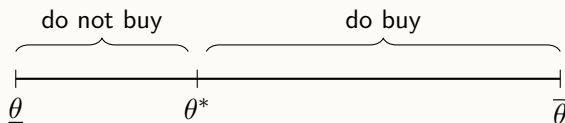


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No extensive margin:



Optimal allocation  $Q_P(\theta)$  maximizes pointwise:

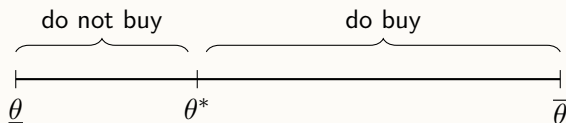
$$1 - Q_P(\theta) = \frac{1}{\gamma} \frac{1 - F(\theta)}{f(\theta)}$$

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$$\int_{\theta^*}^{\bar{\theta}} \left\{ \underbrace{\theta \underline{Q} - \frac{\gamma}{2}(1 - Q(\theta))^2 - u(\underline{Q}; \theta)}_{\text{surplus}} - \underbrace{(Q(\theta) - \underline{Q}) \frac{1 - F(\theta)}{f(\theta)}}_{\text{information rent}} \right\} f(\theta) d\theta$$

No extensive margin:

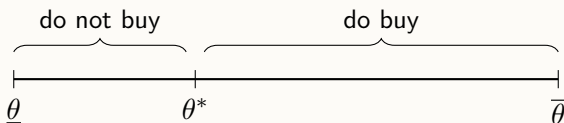


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No extensive margin:



Differentiating the optimal allocation:

$$\dot{Q}_P(\theta) = \frac{1}{\gamma} + (1 - Q_P(\theta)) \frac{f'(\theta)}{f(\theta)}$$

## Horizontal Differentiation: The Extensive Margin

Thought experiment: lower prices uniformly. Effect?

Higher indirect utility  $v(\theta) \Leftrightarrow$  higher demand

$$v(\theta) = \sup_{\tilde{\theta}} U(Q(\tilde{\theta}), \theta) - p(\tilde{\theta})$$

Demand from types  $\theta$  is proportional to:

$$\hat{M}(\theta) = M\left(\underbrace{v(\theta) - U(\underline{Q}, \theta)}_{\text{own added utility} \nearrow}, \underbrace{\bar{v}(\theta) - U(\underline{Q}, \theta)}_{\text{competitor added utility} \searrow}\right)$$

where  $\bar{v}(\theta)$  is indirect utility offered by competitors.

*"We are struck by the fact that, in this instance, econometric practice is ahead of the theoretical literature."*

— Rochet & Stole (2002)



## Example: Multinomial Logit

### Example (Multinomial Logit)

$M(\cdot)$  derives from a random utility model. Type  $\theta$  utility is

$$U(Q, \theta) - p + \xi$$

where  $\xi$  is a Gumbel-distributed error with scale  $\delta > 0$ .

Then  $M(\cdot)$  is multinomial logit with  $N$  competitors:

$$M(\cdot) = \frac{\exp\left(\frac{v(\theta) - U(\underline{Q}, \theta)}{\delta}\right)}{1 + \exp\left(\frac{v(\theta) - U(\underline{Q}, \theta)}{\delta}\right) + (N - 1) \exp\left(\frac{\bar{v}(\theta) - U(\underline{Q}, \theta)}{\delta}\right)}$$

Parameter  $\delta$  governs horizontal differentiation:  $\delta \rightarrow 0$  gives pure monopoly,  $\delta \rightarrow \infty$  with  $N$  fixed gives captive demand

## Example: Hotelling on the Line

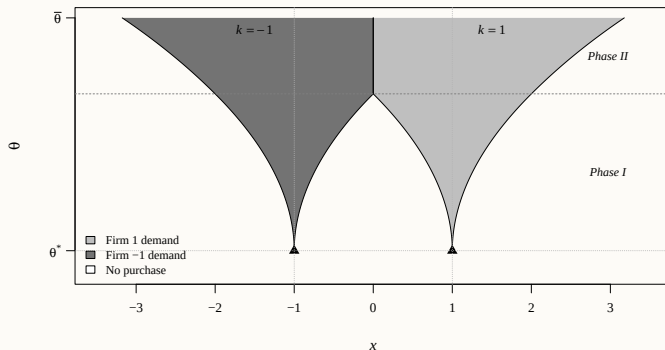
### Example (Hotelling on the Line)

Two firms  $k \in \{-1, 1\}$ . Horizontal types  $x \sim H$  distributed on  $(-\infty, \infty)$  (e.g., Laplace) with transport cost  $\tau$ . Type  $(\theta, x)$  utility from firm  $k$  is

$$U(Q, \theta) - p - \tau|x - k|.$$

Then  $M(\cdot)$  equals the mass of types  $x$  preferring firm  $k$  over the outside option or competitors.

# Hotelling: Participation Regions

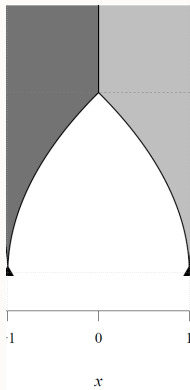


## Example (Hotelling on the Line)

Seller  $k$ 's demand share is

$$M(\cdot) \equiv \begin{cases} H\left(1 + \frac{v^k(\theta) - U(\underline{Q}; \theta)}{\tau}\right) - H\left(\frac{v^{-k}(\theta) - v^k(\theta)}{2\tau}\right) & \text{if } \theta \in (\hat{\theta}, \bar{\theta}] \\ H\left(1 + \frac{v^k(\theta) - U(\underline{Q}; \theta)}{\tau}\right) - H\left(1 - \frac{v^k(\theta) - U(\underline{Q}; \theta)}{\tau}\right) & \text{if } \theta \in [\theta^{*k}, \hat{\theta}] \end{cases}$$

## Hotelling: Uniform Specification



### Example (Hotelling on the Unit Interval)

With uniform horizontal types on  $[-1, 1]$ :

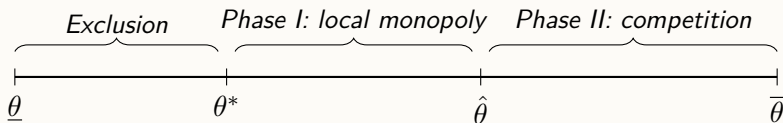
$$M(\cdot) \equiv \begin{cases} \frac{1}{2} + \frac{v(\theta) - \bar{v}(\theta)}{2\tau} & \text{if } \theta \in (\hat{\theta}, \bar{\theta}] \\ \frac{v(\theta) - U(\underline{Q}; \theta)}{\tau} & \text{if } \theta \in [\theta^*, \hat{\theta}] \end{cases}$$

# Uniform Hotelling

# Hotelling on the Unit Interval

Two insurers at  $k \in \{-1, 1\}$ , horizontal types  $x \sim \text{Uniform}[-1, 1]$ , transport cost  $\tau$ .

**Three market regions:**



**Per-type profit:**

$$\pi(\theta) = t(\theta) - \theta(Q(\theta) - \underline{Q}) = -v(\theta) + \theta\underline{Q} - \frac{\gamma}{2}(1 - Q(\theta))^2$$

# Monopoly versus Oligopoly

**Monopoly** (collusion): controls both  $v$  and  $\bar{v}$ ; sets  $v = \bar{v}$ :

$$\int_{\theta^*}^{\hat{\theta}} \pi(\theta) \cdot \frac{v(\theta) - U(\underline{Q}; \theta)}{\tau} d\theta + \int_{\hat{\theta}}^{\bar{\theta}} \pi(\theta) \cdot \frac{1}{2} d\theta$$

---

**Oligopoly**: takes  $\bar{v}(\theta)$  as given:

$$\int_{\theta^*}^{\hat{\theta}} \pi(\theta) \cdot \frac{v(\theta) - U(\underline{Q}; \theta)}{\tau} d\theta + \int_{\hat{\theta}}^{\bar{\theta}} \pi(\theta) \cdot \left( \frac{1}{2} + \frac{v(\theta) - \bar{v}(\theta)}{2\tau} \right) d\theta$$

---

Phase I objectives are identical. The regimes differ only in Phase II.

This is a **calculus of variations** problem: the state  $v(\theta)$  satisfies  $\dot{v}(\theta) = Q(\theta)$ , so the Euler–Lagrange equation characterizes the optimal  $Q(\theta)$ .

# Equilibrium Coverage

Euler–Lagrange in each phase and smooth pasting across:

## Monopoly (collusion)

*Phase I* (local monopoly):

$$\dot{Q}_M(\theta) = \frac{1}{\gamma}$$

*Phase II* (competition):

$$\dot{Q}_M(\theta) = \frac{1}{\gamma}$$

## Oligopoly

*Phase I* (local monopoly):

$$\dot{Q}_O(\theta) = \frac{1}{\gamma}$$

*Phase II* (competition):

$$\dot{Q}_O(\theta) = \frac{1}{\gamma} - \frac{1}{\gamma} \frac{\pi_O(\theta)}{\tau}$$

The regimes differ *only* in Phase II, via the **business stealing term**:

$\pi(\theta) > 0$ : undercutting  $\Rightarrow$  flatten distortions

$\pi(\theta) < 0$ : lemon-dropping  $\Rightarrow$  steepen distortions



## Result 1: Negative Profit at the Top

### Proposition

*There exists  $\theta^{**} > \hat{\theta}$  so that  $\pi_O(\theta) < 0$  for all  $\theta \in (\theta^{**}, \bar{\theta})$ .*

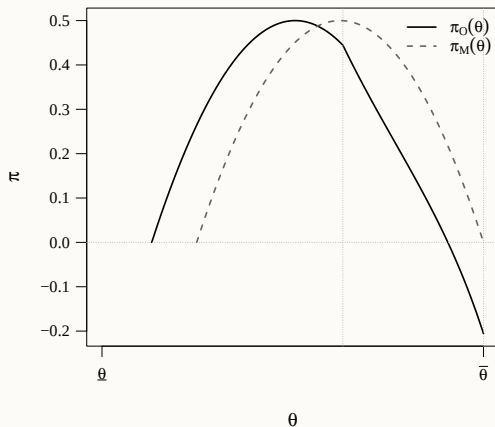
Comprehensive coverage is sold at a loss, cross-subsidized by intermediate contracts.

Note: Suffering losses on comprehensive coverage is a **choice**.

- ▶ The seller could have increased premia on comprehensive coverage
- ▶ By IC, this would require to also raise premia on lesser coverage levels
- ▶ Trade-off: Accept losses at the top to attract greater demand at the profitable bottom

Under private values this cannot arise: higher-valuation buyers are always more profitable.

## Per-Type Profit: Illustration



Per-type profit  $\pi(\theta)$  is positive for intermediate types and negative near  $\bar{\theta}$ . The crossing point  $\theta^{**}$  marks where lemon-dropping begins.

## Proof of Negative Profit at the Top

Define the auxiliary function:

$$g(\theta) \equiv \frac{\pi(\theta)}{\gamma} - \underbrace{(1 - Q(\theta))(Q(\theta) - \underline{Q})}_{\text{"coverage gap"}}$$

**Step 1:** At  $\theta^*$ :  $Q(\theta^*) = \underline{Q}$ ,  $v(\theta^*) = U(\underline{Q}; \theta^*) \Rightarrow g(\theta^*) = 0$ .

**Step 2:** Show  $g(\theta) \leq 0$  for all  $\theta > \theta^*$  by contradiction.

- If  $g$  goes positive,  $\exists \theta'$  with  $g(\theta') \geq 0$  and  $g'(\theta') > 0$ .
- Compute:  $g'(\theta') = (\gamma \dot{Q}(\theta') - 1)(Q(\theta') - \underline{Q}) \Rightarrow \dot{Q}(\theta') > \frac{1}{\gamma}$ .
- But the ODE gives (substituting  $\pi = \gamma g + \gamma(1 - Q)(Q - \underline{Q})$ ):

$$\dot{Q}(\theta') = \frac{1}{\gamma} - \frac{1}{\gamma\tau} \left[ \underbrace{\gamma g(\theta')}_{\geq 0} + \underbrace{\gamma(1 - Q)(Q - \underline{Q})}_{\geq 0} \right] \leq \frac{1}{\gamma}$$

— contradiction.

**Step 3:** At  $\bar{\theta}$ :  $Q(\bar{\theta}) = 1$ , so the coverage gap vanishes:

$$0 > g(\bar{\theta}) = \frac{\pi(\bar{\theta})}{\gamma} \Rightarrow \pi(\bar{\theta}) < 0. \quad \square$$

## Result 2: Competition Exacerbates Distortions at the Top

Since  $\pi_O(\theta) < 0$  for  $\theta$  near  $\bar{\theta}$ , the oligopoly ODE gives:

$$\dot{Q}_O(\theta) = \frac{1}{\gamma} - \underbrace{\frac{\pi_O(\theta)}{\gamma\tau}}_{>0 \text{ since } \pi < 0} > \frac{1}{\gamma} = \dot{Q}_M(\theta)$$

### Proposition

For all  $\theta \in (\theta^{**}, \bar{\theta})$ :

$$\dot{Q}_O(\theta) > \dot{Q}_M(\theta) \quad \text{and} \quad Q_O(\theta) < Q_M(\theta).$$

*Competition reduces the set of types who purchase near-comprehensive coverage.*

### **This directly contrasts with Yang & Ye (2008):**

In their private-values model, high types are *profitable*. Competition incentivizes firms to offer more generous coverage to steal these valuable buyers. With adverse selection, competition works in reverse: driving away loss-making buyers increases profit at the rival's expense.

## Result 3: Inclusion at the Bottom

### Proposition

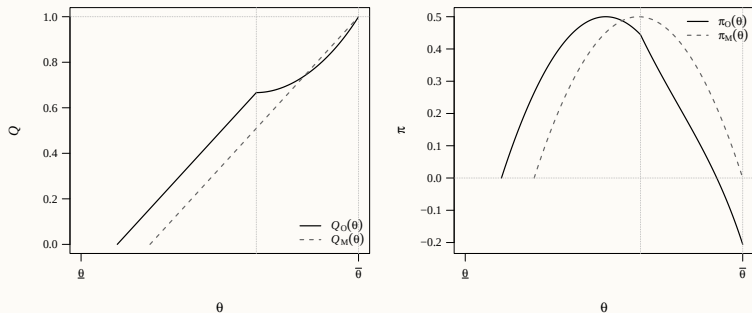
$\theta_O^* \leq \theta_M^*$ : *oligopoly leads to less exclusion than monopoly.*

**Single-crossing:** The coverage assignments  $Q_O(\theta)$  and  $Q_M(\theta)$  cross exactly once. Below the crossing point, oligopoly provides more generous coverage. Above it, monopoly does.

**Welfare comparison is ambiguous:**

- ▶ At the bottom: oligopoly is better (less exclusion)
- ▶ At the top: monopoly is better (less coverage distortion)
- ⇒ Overall ranking depends on welfare weights across buyer types

# Monopoly versus Oligopoly: Illustration



Collusion = monopoly with extensive margin. Single-crossing of  $Q_M$  and  $Q_O$ : oligopoly improves inclusion at the bottom but steepens distortions at the top.

The derived ODEs characterize the solution provided

$\sqrt{2\tau/\gamma} \geq \frac{2}{3}(1 - \underline{Q})$ . This ensures  $\dot{Q}(\theta) \geq 0$  throughout (no bunching).

# General Distributions

## Two Market Structures

**Monopoly**: controls both  $v(\theta)$  and  $\bar{v}(\theta)$ ; sets  $v = \bar{v}$ :

$$M(\cdot) = M(\textcolor{red}{v}(\theta) - U(\underline{Q}, \theta), \textcolor{red}{v}(\theta) - U(\underline{Q}, \theta))$$

**Oligopoly**: controls  $v(\theta)$ , takes  $\bar{v}(\theta)$  as given:

$$M(\cdot) = M(\textcolor{red}{v}(\theta) - U(\underline{Q}, \theta), \textcolor{blue}{\bar{v}}(\theta) - U(\underline{Q}, \theta))$$

In both regimes, the insurer faces the same **optimal control problem**, differing only in how  $M(\cdot)$  responds to  $v(\theta)$ .



# Optimal Pricing with Extensive Margin

The insurer solves an **optimal control** problem:

$$\max_{(v, Q), z} \int_{\underline{\theta}}^{\bar{\theta}} \left[ \underbrace{\theta Q(\theta) - \frac{\gamma}{2}(1 - Q(\theta))^2 - v(\theta)}_{\text{price } p(\theta)} - \underbrace{\theta(Q(\theta) - \underline{Q})}_{\text{cost}} \right] f(\theta) \underbrace{M(\cdot)}_{\text{extensive margin}} d\theta$$

s.t.  $(v, Q)$  is absolutely continuous and  $z$  is measurable

$$\dot{v}(\theta) = Q(\theta) \quad \text{and} \quad \dot{Q}(\theta) = z(\theta) \quad \text{a.e.}$$

$$z(\theta) \geq 0 \quad \text{a.e.}$$

$$v(\underline{\theta}) \geq U(\underline{Q}; \theta), \quad Q(\underline{\theta}) \geq \underline{Q}, \quad v(\bar{\theta}) \text{ is free and } Q(\bar{\theta}) \leq 1$$

*Constraints: binding local ICs + participation constraint at the bottom.*

States:  $(v, Q)$ .      Control:  $z = \dot{Q} \geq 0$ .      Costates:  $(\lambda_v, \lambda_Q)$ .

# Pontryagin's Maximum Principle

**Hamiltonian** (Clarke 2013, Theorem 22.20):

$$\mathcal{H} = - \underbrace{[U(Q; \theta) - v - \theta(Q - \underline{Q})]}_{\pi(\theta)} \hat{M}(\theta) f(\theta) + \lambda_v Q + \lambda_Q z$$

where  $\hat{M}(\theta) = M(v(\theta) - U(\underline{Q}; \theta), \bar{v}(\theta) - U(\underline{Q}; \theta))$ .

**Maximum condition** ( $\mathcal{H}$  is linear in the control  $z$ ):

$$\lambda_Q(\theta) \leq 0 \text{ for all } \theta, \quad \text{with equality whenever } \dot{Q}(\theta) > 0$$

This yields the key distinction:

- **Separating** ( $\dot{Q} > 0$ ):  $\lambda_Q(\theta) = 0 \Rightarrow \dot{\lambda}_Q = 0$  and  $\ddot{\lambda}_Q = 0$
- **Bunching** ( $\dot{Q} = 0$ ):  $\lambda_Q(\theta) \leq 0$

**Adjoint equations**  $\dot{\lambda}_Q = -D_Q \mathcal{H}$  and  $\dot{\lambda}_v = -D_v \mathcal{H}$

The equation  $\ddot{\lambda}_Q(\theta) = 0$  yields the **characterizing ODE** for coverage.

# General Case: Monopoly

## Regularity Conditions

Log-concave demand that responds concavely to symmetric utility increases:

### Assumption

- (i)  $w \mapsto \frac{d_1 M(w, w) + d_2 M(w, w)}{M(w, w)}$  and  $w \mapsto \frac{d_1 M(w, w)}{M(w, w)}$  are non-increasing
- (ii)  $d_1 M(w, w) + d_2 M(w, w) \geq 0$
- (iii)  $d_{12}^2 M(w, w) + d_{22}^2 M(w, w) \leq 0$

Type distribution  $f(\theta)$  well-behaved (e.g., exponential, uniform):

### Assumption

- (i)  $\frac{f'(\theta)}{f(\theta)} \in \left(-\frac{1}{\gamma}, 0\right]$
- (ii)  $\theta \mapsto \frac{f'(\theta)}{f(\theta)}$  is non-decreasing

Note:  $f(\theta)$  uniform in Rochet & Stole (2002) and Yang & Ye (2008).

# Optimal Monopoly Coverage

## Theorem

If the profit-maximizing  $\theta \mapsto Q(\theta)$  is absolutely continuous, there exists some  $\theta^* \in [\underline{\theta}, \bar{\theta}]$  so that:

$$Q(\theta) = \underline{Q} + q(\theta) \quad \begin{cases} = Q^* \in [\underline{Q}, 1) & \text{if } \theta \in [\underline{\theta}, \theta^*) \\ \text{solves ODE} & \text{if } \theta \in [\theta^*, \bar{\theta}] \\ = 1 & \text{if } \theta = \bar{\theta}, \end{cases}$$

where, within the separating interval  $[\theta^*, \bar{\theta}]$ , total coverage solves

$$\dot{Q}(\theta) = \underbrace{(1 - Q(\theta)) \frac{f'(\theta)}{f(\theta)} + \frac{1}{\gamma}}_{\text{pure monopoly}} - \underbrace{\frac{d_1 M(\cdot) + d_2 M(\cdot)}{M(\cdot)} \frac{1}{q(\theta)} \left[ \frac{1}{2} Q^2 - \frac{v(\theta) - \theta Q}{\gamma} - \frac{1}{2} + (1 - Q) \underline{Q} \right]}_{\text{extensive margin}}$$

# The Extensive Margin Term

Pure monopoly ODE:

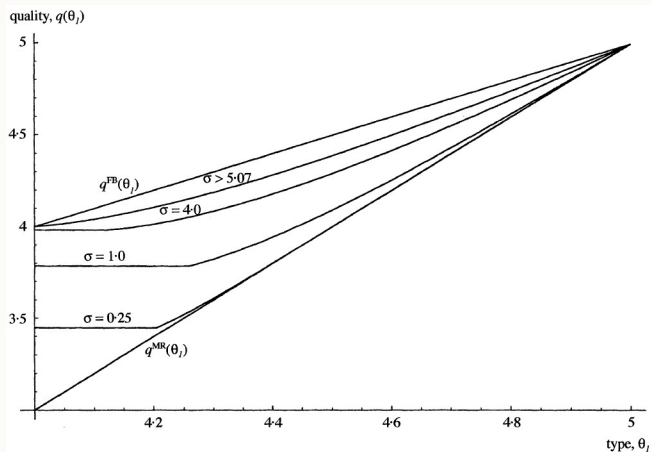
$$\dot{Q}_P(\theta) = (1 - Q) \frac{f'(\theta)}{f(\theta)} + \frac{1}{\gamma}$$

With an extensive margin, the ODE acquires an additional term:

$$\begin{aligned} \dot{Q}_M(\theta) = & \underbrace{(1 - Q) \frac{f'(\theta)}{f(\theta)} + \frac{1}{\gamma}}_{\dot{Q}_P(\theta)} \\ & - \underbrace{\frac{d_1 M + d_2 M}{M}}_{\geq 0} \frac{1}{q} \underbrace{\left[ \frac{1}{2} Q^2 - \frac{v - \theta Q}{\gamma} - \frac{1}{2} + (1 - Q) Q \right]}_{\equiv g(\theta)} \end{aligned}$$

So  $\dot{Q}_M \geq \dot{Q}_P$  according to  $g(\theta) \leq 0$ .

# Rochet & Stole (2002)



# Extensive Margin Exacerbates Distortions

## Proposition

*Under Theorem 1's assumptions and when  $Q(\theta)$  is supported on  $[Q, 1]$ : the extensive margin exacerbates distortions to coverage:*

$$1 = Q_{fb}(\theta) \geq Q_P(\theta) \geq Q_M(\theta)$$

*and the latter inequality is strict for types where  $f(\theta)$  is non-uniform.*



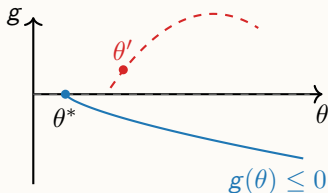
**Proof:**  $g(\theta) \leq 0$

Rewrite using  $v(\theta) = v(\theta^*) + \int_{\theta^*}^{\theta} Q(\vartheta) d\vartheta$  and  $w_0 \equiv v(\theta^*) - U(\underline{Q}; \theta^*) \geq 0$ :

$$g(\theta) = \frac{1}{2}(Q - \underline{Q})^2 - \frac{1}{\gamma} \int_{\theta^*}^{\theta} (Q(\vartheta) - \underline{Q}) d\vartheta - \frac{w_0}{\gamma}$$

Initially  $g(\theta^*) = 0$  since  $Q(\theta^*) = \underline{Q}$  and  $p(\theta^*)$  by assumption.

Suppose  $g(\theta') > 0$  for some  $\theta' > \theta^*$ . Then  $\exists \theta'$  with  $g(\theta') \geq 0$  and  $g'(\theta') > 0$ .



Compute:  $g'(\theta') = (Q - \underline{Q})(\dot{Q} - \frac{1}{\gamma}) > 0 \Rightarrow \dot{Q}(\theta') > \frac{1}{\gamma}$ .

But from the ODE,  $g(\theta') \geq 0$  implies:

$$\begin{aligned}\dot{Q}(\theta') &= (1 - Q) \frac{f'}{f} + \frac{1}{\gamma} - \underbrace{\frac{d_1 M + d_2 M}{M}}_{\geq 0} \frac{1}{q} \underbrace{g(\theta')}_{\geq 0} \\ &\leq (1 - Q) \underbrace{\frac{f'}{f}}_{\leq 0} + \frac{1}{\gamma} \leq \frac{1}{\gamma}\end{aligned}$$

Contradiction.



## Negative Profit at the Top

### Proposition

*Under Theorem 1's assumptions and  $Q$  supported on  $[\underline{Q}, 1]$ :  $\pi(\bar{\theta}) \leq 0$ , strictly if  $f'(\bar{\theta}) < 0$ . If  $\frac{f'(\theta)}{f(\theta)}$  is constant, profit is downcrossing: positive below some threshold and negative above it.*

Note: Monopoly losses unless uniform types.

**Proof.** Recall  $g(\theta) \leq 0$ . Rewrite as:

$$g(\theta) = \frac{\pi(\theta)}{\gamma} - (1 - Q(\theta))(Q(\theta) - \underline{Q})$$

At  $\bar{\theta}$ :  $Q(\bar{\theta}) = 1$ , so the coverage gap vanishes:

$$0 \geq g(\bar{\theta}) = \frac{\pi(\bar{\theta})}{\gamma} \quad \Rightarrow \quad \pi(\bar{\theta}) \leq 0 \quad \square$$

Same cross-subsidization logic as Hotelling: losses at the top are a *choice*, sustained by IC to retain profitable demand at the bottom.

# General Case: Oligopoly

## Additional Assumptions for Oligopoly

To rule out bunching under competition, we strengthen prior assumptions:

Curvature of the extensive margin w.r.t. competitor utility:

Assumption

$$(iv) \frac{d_{12}^2 M(\cdot) + d_{22}^2 M(\cdot)}{d_2 M(\cdot)} (1 - \underline{Q}) < \frac{1}{\gamma}$$

No thin high-risk tails:

Assumption

$$(v) \frac{f'(\theta)}{f(\theta)} \theta > -\frac{1}{\gamma}$$

MNL: Assumption 3 requires  $\delta$  or  $N$  sufficiently large. Assumption 4 is mild — rules out type densities declining too steeply near the top.

# Equilibrium Oligopoly Coverage

## Theorem

In any symmetric oligopoly equilibrium where  $\theta \mapsto Q(\theta)$  is absolutely continuous there exists some  $\theta^* \in [\underline{\theta}, \bar{\theta}]$  s.t.:

$$Q(\theta) = \underline{Q} + q(\theta) \quad \begin{cases} = Q^* \in [\underline{Q}, 1) & \text{if } \theta \in [\underline{\theta}, \theta^*) \\ \text{solves ODE} & \text{if } \theta \in [\theta^*, \bar{\theta}] \\ = 1 & \text{if } \theta = \bar{\theta}, \end{cases}$$

where total coverage solves

$$\begin{aligned} \dot{Q}(\theta) = & \underbrace{(1 - Q) \frac{f'(\theta)}{f(\theta)} + \frac{1}{\gamma}}_{\text{pure monopoly}} \underbrace{- \frac{d_1 M + d_2 M}{M} \frac{1}{q} \left[ \frac{\pi}{\gamma} - (1 - Q)q \right]}_{\text{extensive margin (both regimes)}} \\ & \underbrace{- \frac{d_2 M}{M} \left[ \frac{v - \theta \underline{Q}}{\gamma} + \frac{1}{2} (1 - Q)^2 \right]}_{\text{business stealing (oligopoly only)}} \end{aligned}$$

# The Role of Per-Type Profit

Rewrite the system in terms of  $Q(\theta)$  and  $\pi(\theta)$ :

$$\begin{aligned}\dot{Q}(\theta) = & \underbrace{(1 - Q) \frac{f'}{f} + \frac{1}{\gamma}}_{\text{pure monopoly}} - \underbrace{\frac{d_1 M + d_2 M}{M} \cdot \frac{1}{q} \left[ \frac{\pi}{\gamma} - (1 - Q) q \right]}_{\text{extensive margin (both regimes)}} \\ & - \underbrace{\frac{d_2 M}{M} \cdot \frac{\pi}{\gamma}}_{\text{business stealing}}\end{aligned}$$

Since  $d_2 M \leq 0$ , the business-stealing term enters with the **sign of**  $\pi(\theta)$ :

- ▶  $\pi(\theta) > 0$ : competition *alleviates* distortions (undercutting)
- ▶  $\pi(\theta) < 0$ : competition *exacerbates* distortions (lemon-dropping)

We already know:  $\pi(\bar{\theta}) < 0$ . So lemon-dropping dominates near the top.

## Competition Exacerbates Distortions at the Top

For the general proof, we need one more assumption:

### Assumption

$$(vi) \quad w \mapsto \frac{d_1 M(w, w) + d_2 M(w, w)}{M(w, w)} \cdot w \text{ is non-decreasing}$$

MNL: satisfied when  $\delta \geq w_{\max}$ .

### Proposition

*If  $v_O(\theta) \geq v_M(\theta)$  (oligopoly provides greater consumer rents), then oligopoly results in greater distortions at the top:*

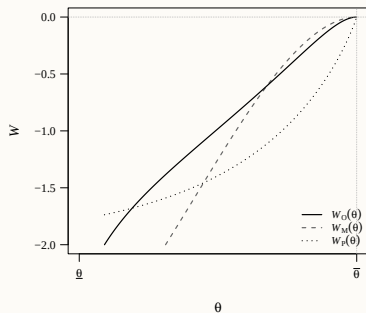
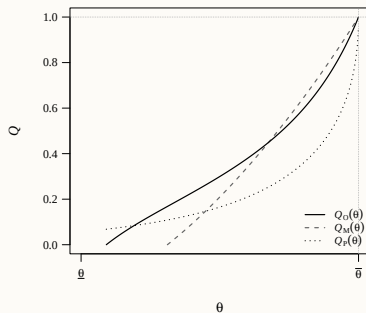
$$\dot{Q}_O(\bar{\theta}) \geq \dot{Q}_M(\bar{\theta})$$

*with strict inequality unless  $f$  is uniform and  $d_2 M = 0$ .*

Coverage distortions at the top are **greater under oligopoly**: competition makes it harder for high-risk types to obtain near-comprehensive coverage.



# MNL: Monopoly versus Oligopoly



Coverage and per-type welfare: monopoly versus oligopoly in the random utility model. Oligopoly features less exclusion at the bottom but greater coverage distortions and lower welfare at the top.

## Proof: Distortions at the Top (General Case)

Know:  $v_O(\bar{\theta}) \geq v_M(\bar{\theta})$ , i.e.,  $w_O(\bar{\theta}) \geq w_M(\bar{\theta})$ .

Take differences of the monopoly and oligopoly ODEs at  $\bar{\theta}$ :

$$\begin{aligned}\dot{Q}_O(\bar{\theta}) - \dot{Q}_M(\bar{\theta}) &\geq \underbrace{\frac{d_1 M + d_2 M}{M} \Big|_{w_O} \cdot \frac{w_O}{\gamma} - \frac{d_1 M + d_2 M}{M} \Big|_{w_M} \cdot \frac{w_M}{\gamma}}_{\geq 0 \text{ by Assumption 5}} \\ &\quad + \underbrace{\frac{d_2 M}{M} \Big|_{w_O} \cdot \frac{\pi_O}{\gamma}}_{\geq 0 \text{ since } d_2 M \leq 0, \pi_O \leq 0}\end{aligned}$$

Both terms are non-negative; the second is strictly positive when  $\pi(\bar{\theta}) < 0$ .  $\square$

# Equilibrium Existence

## Equilibrium Existence

So far: Pontryagin conditions characterize *candidate* equilibria.

Are they globally optimal? Under perfect competition, the answer is no: Rothschild & Stiglitz (1976) non-existence.

**Approach:** Arrow–Mangasarian sufficiency

If the **maximized Hamiltonian** is globally concave in the states  $(v, Q)$ , then any solution to the Pontryagin conditions is a global optimum — hence an equilibrium.

**Concavity requires:** for all deviations  $(\tilde{Q}, w, \theta)$ ,

$$2 - \pi_{\Delta} \frac{d_{11}^2 M}{d_1 M} \geq \gamma(1 - \tilde{Q})^2 \frac{d_1 M}{M}$$

where  $\pi_{\Delta}(\tilde{Q}, w, \theta) = \underbrace{U(\tilde{Q}; \theta) - U(\underline{Q}; \theta)}_{\text{surplus from coverage } \tilde{Q}} - w - \theta(\tilde{Q} - \underline{Q})$

is per-type profit at an arbitrary *deviation*  $(\tilde{Q}, w)$  from the candidate equilibrium.

## Existence under Multinomial Logit

Under MNL, demand derivatives scale as  $1/\delta$ .

The concavity condition reduces to a simple inequality:

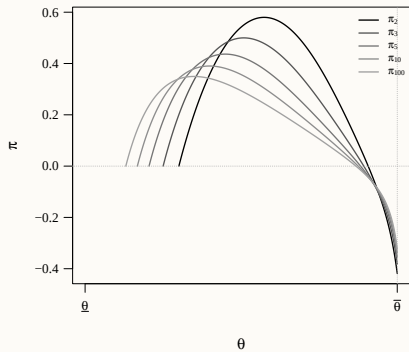
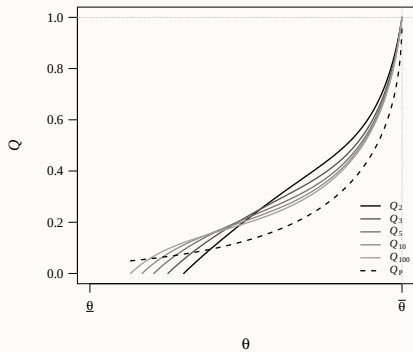
$$\delta \geq w_{\max} \equiv \bar{\theta}(1 - \underline{Q}) + \frac{\gamma}{2}(1 - \underline{Q})^2$$

where  $w_{\max}$  is the maximal added utility (at full coverage for the highest type).

### Implications:

- ▶ Equilibrium exists for any number of firms  $N \geq 2$
- ▶ In particular, existence holds in the limit  $N \rightarrow \infty$
- ▶ Sufficient differentiation ( $\delta$  large) ensures insurers can only *gradually* shed buyers onto competitors — ruling out profitable large deviations

This partially addresses the Rothschild–Stiglitz non-existence concern: horizontal differentiation restores existence of pure-strategy equilibria.



When  $N \rightarrow \infty$

# Conclusion and Policy Implications (tbc)

## Policy Implications

### Increasing basic coverage $\underline{Q}$ :

- ▶ Greater coverage among insured
- ▶ But: fosters greater *opt-out* among low- $\theta$  types — with higher  $\underline{Q}$ , the surplus from private top-up shrinks, so more buyers exit the private market, worsening adverse selection

**Public reinsurance:** a more targeted instrument.

Key insight: all our results are driven by  $\pi(\bar{\theta}) < 0$ .

Public reinsurance absorbs losses on high-cost enrollees  $\Rightarrow$  transforms the **sign of profit at the top**.

Once comprehensive coverage is no longer loss-making:

- ▶ Overcutting incentives vanish
- ▶ Standard competitive forces resume: undercutting, not lemon-dropping

Kim & Li (2026): state-level public reinsurance programs reduce health insurance premiums by 15%, with a **pass-through rate exceeding one**.



## Reinsurance: Empirical Motivation

Kim & Li (2026): state-level public reinsurance programs reduce health insurance premiums by 15%, with a **pass-through rate exceeding one**.

For every dollar the government spends on reinsurance, premiums fall by more than a dollar.

Their explanation: reduced insurer risk exposure.

### **Our complementary mechanism:**

Reinsurance does not merely offset costs — it switches the competitive regime from lemon-dropping to undercutting.

A pass-through rate  $> 1$  is a natural consequence in spite of market power.

To be developed: formal reinsurance analysis within our framework, welfare comparisons, and optimal reinsurance design.

# Microfoundation of Utility Function

[Back to main presentation](#)

**Setup:** CARA utility with coefficient  $\alpha$  and normal health shocks  $L \sim N(\theta, \sigma^2)$ :

$$\begin{aligned}\bar{U}(Q, \theta) - p &= \mathbb{E} \left[ 1 - \exp \left( -\alpha(W - L(1 - Q) - p) \right) \right] \\ &= 1 - \mathbb{E} \left[ \exp \left( \alpha L(1 - Q) \right) \right] \exp \left( -\alpha W + \alpha p \right) \\ &= 1 - \exp \left( \alpha \theta (1 - Q) + \frac{\alpha^2 \sigma^2 (1 - Q)^2}{2} \right) \exp \left( -\alpha W + \alpha p \right)\end{aligned}$$

**Apply monotone transformation**  $x \mapsto -\frac{1}{\alpha} \log(-x)$ :

$$\begin{aligned}U(Q; \theta) - p &= -\theta(1 - Q) - \underbrace{\alpha \sigma^2}_{\gamma} \frac{(1 - Q)^2}{2} + W - p \\ &= \theta Q - \frac{\gamma}{2} (1 - Q)^2 - p + \text{const.}\end{aligned}$$